The distributive law

1 The student should multiply the second part of the sum, 2, by the number outside the brackets, 6 to get 12 as the second term on the right hand side.

2

a $5s + 35$

b $45 + 9w$

c $3r - 18$

d $7y + 56$

e $-8t - 80$

f $-8c + 40$

g $-y - 2$

h $16s + 20$

i $-80 - 30r$

j $16t + 24$

k $pq + 4p$

l $9u - 9v$

m $-2m + 3n$

n $-11rs - 13rt$

o $8r + 12s + 20t$

p $24m + 18n + 30p$

3

a $12mn + 18m$

b $9v^2 + 54$

c $r^2 + 7r$

d $24m^2 + 42m$

e $-12a^2 - 54a$

f $4wy^2 + 4w^2y$

g $16pq - 12p + 24pr$

h $2f^2 - 14fg - 12fh$

i $-10y^2 - 70$

j $-9 + s^3$

k $rs^2 + 8r$

l $-30 - 80u^3$

m $16s^5t^4 - 36s^5$

n $12p^3q^5 - 54p^3$

o $9f^2 - 36fg - 54fh$

p $3uv^2 + 3u^2v$

4

a $4x + 30$

b $15x + 27$

c $3x + 17$

d $18x + 32$

e $36x - 11$

f $-11x + 24$

g $-x - 15$

h $-40x + 47$

i $10y - 49$

j $20x - 52$

5

a $x^2 + 12x + 72$

b $15y^2 - 39y$

c $10x - 33y$

d $12y + 93$

e $13x - 16$

f $4y + 44$

g $-11x - 26$

h $17a + 20$

i $14u^2 + 20uv$

j $62w^2 + 34wz$

Applications

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7 $(16r - 24)$ representatives

8 $(10 + 15n^4)$ boxes

9 $(16s^2t + 16st^2)$ units^2

10 $(u^2 + 8u)$ m^2

Factorisation

11

a Yes	**b** No	**c** Yes	**d** Yes
e No	**f** Yes		

12

a 2	**b** 3	**c** 3	**d** 2
e 7	**f** $4x$	**g** 5	**h** 6
i 3			

13

a 2	**b** 5	**c** 3	**d** 2
e $v + 3$	**f** $a - 4$	**g** $8a + 3$	**h** $2w - 9$
i 4	**j** 5	**k** 6	**l** $r + 7$
m $t - 2$	**n** $9u + 2$		

14

a $6(v + 5)$	**b** $6(a - 8)$	**c** $5(9t - 8)$	**d** $2(m + 5n)$
e $6(9 - 10fg)$	**f** $8(g - 9h)$	**g** $3(3f + 5g)$	**h** $3(w + 5)$
i $2(c - 9)$	**j** $5(8c - 7)$	**k** $6(w + 5y)$	**l** $2(7y + 2w)$
m $2(2a - 7b)$	**n** $4(8 - 9jk)$	**o** $2(4jkt - 9)$	**p** $f(3g - 10h)$
q $rw(3t - 8)$	**r** $2rt(2 - 3r)$	**s** $4wy(4 - 9w)$	

15 $2w - 10z + 8y = 2(w - 5z + 4y)$

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16

a $4(2g+7h-5)$

b $4(w+z+y)$

c $4(u-3v+5w)$

d $5(2m-5n-3p)$

e $3(m-2n+4p)$

f $5(2g-7h+3)$

g $z(w-y-k)$

h $5r(12-5t-7rt)$

i $4m(3-8n-5mn)$

j $4z(4-8y-6y^2)$

k $5rt(5r-3t-9rt)$

Factorisation of negatives

17

a $-(4r+3s)$

b $-7(r+s)$

c $-(2+9r)$

d $-(8x+5y)$

18

a $-8(b+2)$

b $-2(9a-8)$

c $-10(8v+3)$

d $-6(6+x)$

e $-5(5u+7v)$

f $-7(5m-7n)$

g $-2pr(4q+3)$

h $-w^2(8-3y)$

i $-3j^2(8+9k)$

j $-m(8m-9n)$

k $-uv(10u-9v)$

l $-3j^2(7+12k)$

m $-2(s+5)$

n $-8(v^2-7)$

o $-2(6s-5)$

p $-7(4b+3)$

q $-7(9r^2-4)$

r $-4(m+8n)$

s $-7(7y+3w)$

19

a $-9(2p+q+5r)$

b $-5(2r+5s-4t)$

c $-5(4u+5v-9w)$

d $-2(4w+9y+5z)$

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